

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Name: SmartOps

Project Title: AI-Assisted Smart Operations and Service Request Management System

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for **SmartOps**, a web-based operations and service request management application used to manage service requests, task assignments, staff activities, issue resolution, notifications, feedback, reports, analytics, and administrative activities.

The system also includes lightweight AI-assisted features for:

- Service request classification
- Department recommendation
- Priority prediction
- Duplicate request detection

The application is designed as a **MERN-first system**, where MongoDB, Express.js, React.js, and Node.js form the core application, while AI provides intelligent assistance within the existing workflow.

1.2 Scope

SmartOps allows users/employees to register and log in to the system.

Users can create service requests such as:

- IT support issues
- Hardware problems
- Software problems
- Network issues
- Maintenance requests
- Facility issues
- HR-related requests
- General operational requests

Users can provide descriptions, upload supporting images/documents, and provide relevant request information.

Users can:

- Track submitted requests
- View request status
- View assigned staff
- Receive notifications
- Add comments
- Provide feedback after resolution

Staff can:

- View assigned requests
- Accept requests
- Update request status
- Add remarks
- Upload resolution evidence
- Resolve requests
- Manage assigned tasks

Administrators can:

- Manage users
- Manage staff
- Manage departments
- Manage request categories
- Assign requests
- Manage tasks
- Monitor requests
- View feedback
- Generate reports
- View analytics
- Monitor staff performance
- View audit logs

The system provides AI-assisted:

- Request classification
- Department recommendation
- Priority prediction
- Duplicate request detection

2. Stakeholders and User Roles

Role	Description	Main Responsibilities
User / Employee	Person who submits service requests	Registration, login, request creation, file upload, tracking, feedback
Staff / Support Team	Person responsible for handling assigned requests	View requests, update status, add remarks, resolve requests, upload evidence
Administrator	Person responsible for managing the platform	User management, staff management, request assignment, departments, reports, analytics, system monitoring

3. Functional Requirements

3.1 User / Employee Functional Requirements

ID	Requirement	Priority
FR-U01	Allow a user to create an account.	High
FR-U02	Collect the user's name during registration.	High
FR-U03	Collect the user's email address.	High
FR-U04	Collect the user's phone number.	Medium
FR-U05	Collect the user's department.	Medium
FR-U06	Allow the user to create a password.	High
FR-U07	Allow registered users to log in using email and password.	High
FR-U08	Allow users to view and update their profile.	Medium
FR-U09	Allow users to change their password.	Medium
FR-U10	Allow users to create a new service request.	High
FR-U11	Allow users to provide a request title and description.	High
FR-U12	Allow users to select a request category.	High
FR-U13	Allow users to upload images or documents related to a request.	High
FR-U14	Allow users to provide relevant request location information when required.	Medium
FR-U15	Allow users to submit service requests.	High

ID	Requirement	Priority
FR-U16	Allow users to view a list of their submitted requests.	High
FR-U17	Allow users to view request details.	High
FR-U18	Allow users to view the current status of their requests.	High
FR-U19	Allow users to track the progress of their requests.	High
FR-U20	Allow users to view the department or staff assigned to their request.	Medium
FR-U21	Allow users to view request history and status updates.	High
FR-U22	Allow users to add comments to active requests.	Medium
FR-U23	Allow users to receive notifications when request status changes.	Medium
FR-U24	Allow users to receive notifications when requests are assigned or resolved.	Medium
FR-U25	Allow users to provide feedback after a request is resolved.	Medium
FR-U26	Allow users to rate the resolution service.	Medium
FR-U27	Allow users to reset their password.	Medium

3.2 Staff Functional Requirements

ID	Requirement	Priority
FR-ST01	Allow staff to log in using email and password.	High
FR-ST02	Allow staff to view and update their profile.	Medium
FR-ST03	Allow staff to view requests assigned to them.	High
FR-ST04	Allow staff to view request details.	High
FR-ST05	Allow staff to view request location information.	Medium
FR-ST06	Allow staff to view uploaded request attachments.	High
FR-ST07	Allow staff to accept assigned requests.	High
FR-ST08	Allow staff to update request status.	High
FR-ST09	Allow staff to add remarks to requests.	High
FR-ST10	Allow staff to upload images or documents during the resolution process.	Medium

ID	Requirement	Priority
FR-ST11	Allow staff to mark requests as "In Progress".	High
FR-ST12	Allow staff to mark requests as "Resolved".	High
FR-ST13	Allow staff to view previously resolved requests.	Medium
FR-ST14	Allow staff to view request history.	Medium
FR-ST15	Allow staff to receive notifications for newly assigned requests.	Medium
FR-ST16	Allow staff to filter requests by status and priority.	Medium
FR-ST17	Allow staff to view AI-suggested category and priority.	Medium
FR-ST18	Allow staff to view assigned tasks.	High
FR-ST19	Allow staff to update task progress.	High
FR-ST20	Allow staff to mark assigned tasks as completed.	High
FR-ST21	Allow staff to view task deadlines.	Medium

3.3 Administrator Functional Requirements

ID	Requirement	Priority
FR-A01	Allow the Administrator to log in using email and password.	High
FR-A02	Allow the Administrator to add staff accounts.	High
FR-A03	Allow the Administrator to edit staff information.	High
FR-A04	Allow the Administrator to deactivate staff accounts.	High
FR-A05	Allow the Administrator to view and manage user accounts.	High
FR-A06	Allow the Administrator to activate and deactivate user accounts.	High
FR-A07	Allow the Administrator to add, edit, and deactivate departments.	High
FR-A08	Allow the Administrator to add, edit, and deactivate request categories.	High
FR-A09	Allow the Administrator to view all submitted requests.	High
FR-A10	Allow the Administrator to view request details.	High

ID	Requirement	Priority
FR-A11	Allow the Administrator to assign requests to staff.	High
FR-A12	Allow the Administrator to reassign requests to another staff member.	Medium
FR-A13	Allow the Administrator to assign requests to departments.	High
FR-A14	Allow the Administrator to update request priority.	High
FR-A15	Allow the Administrator to monitor request status.	High
FR-A16	Allow the Administrator to manage pending requests.	High
FR-A17	Allow the Administrator to monitor requests that are in progress.	High
FR-A18	Allow the Administrator to monitor resolved requests.	High
FR-A19	Allow the Administrator to manage duplicate requests.	Medium
FR-A20	Allow the Administrator to view user feedback and ratings.	Medium
FR-A21	Allow the Administrator to monitor staff performance.	High
FR-A22	Allow the Administrator to generate request reports.	High
FR-A23	Allow the Administrator to generate reports based on request category.	High
FR-A24	Allow the Administrator to generate reports based on department.	High
FR-A25	Allow the Administrator to generate reports based on request status.	High
FR-A26	Allow the Administrator to generate reports based on priority.	Medium
FR-A27	Allow the Administrator to view resolution statistics.	High
FR-A28	Allow the Administrator to view operational statistics and analytics.	High
FR-A29	Allow the Administrator to monitor request transactions.	High
FR-A30	Allow the Administrator to monitor user activities.	Medium
FR-A31	Allow the Administrator to view system audit logs.	High

3.4 AI-Assisted Functional Requirements

ID	Requirement	Priority
FRAI01	Allow the system to analyze the description of a submitted service request.	High
FRAI02	Allow the system to suggest an appropriate request category using AI.	High
FRAI03	Allow the system to suggest the responsible department based on the request.	Medium
FRAI04	Allow the system to analyze a request and suggest its priority.	High
FRAI05	Allow the system to classify requests into Low, Medium, High, or Critical priority.	High
FRAI06	Allow the system to compare a new request with existing requests.	Medium
FRAI07	Allow the system to identify potentially duplicate requests.	Medium
FRAI08	Allow authorized Administrators to review AI-generated category suggestions.	High
FRAI09	Allow authorized Administrators to accept or modify AI-generated suggestions.	High
FRAI10	Allow the system to store AI-generated classification information along with the request.	Medium

4. Business Rules

ID	Business Rule
BR01	A user must register before accessing user-specific features.
BR02	Authentication shall be required before accessing role-specific dashboards.
BR03	Users shall only be able to access functionality permitted by their assigned role.
BR04	A user must have an active account to submit a service request.
BR05	A request must contain the required title and description before submission.
BR06	Only authorized Administrators should be able to assign requests to staff.
BR07	Only authorized staff should be able to update requests assigned to them.
BR08	Request status changes shall be recorded in request history.
BR09	A request marked as resolved should contain appropriate resolution information.
BR10	Users should be able to provide feedback only after a request has been resolved.
BR11	AI-generated category and priority suggestions should be reviewable by authorized Administrators.

ID Business Rule

BR12	AI suggestions shall not prevent an Administrator from modifying the category or priority.
BR13	Potentially duplicate requests shall be reviewed by an authorized Administrator before being treated as duplicates.
BR14	Only authorized Administrators should be able to manage users, staff, departments, reports, system configuration, and audit logs.
BR15	Important user activities and administrative actions should be recorded in audit logs.

5. Non-Functional Requirements

ID	Category	Requirement
NFR01	Security	Securely authenticate users using email and password.
NFR02	Authorization	Enforce role-based access for Users, Staff, and Administrators.
NFR03	Data Privacy	Protect user, staff, request, and operational information from unauthorized access.
NFR04	Password Security	Store passwords using secure password hashing rather than plain text.
NFR05	Performance	Normal request-management operations should respond within an acceptable time under expected system load.
NFR06	Availability	The system should be available whenever the organization requires operational service management.
NFR07	Usability	The interface should be simple and understandable for all three roles.
NFR08	Reliability	Maintain data consistency for users, requests, assignments, status updates, notifications, tasks, and feedback.
NFR09	Scalability	Support increasing numbers of users, staff, requests, tasks, and transactions.
NFR10	Maintainability	Use a modular architecture that allows independent maintenance of major system features.
NFR11	Compatibility	Work correctly on modern web browsers and desktop/mobile screen sizes.
NFR12	Auditability	Administrative actions and important request transactions should be traceable through audit logs.
NFR13	Data Integrity	Keep user, request, assignment, task, status, feedback, and transaction data accurate and consistent.

ID	Category	Requirement
NFR14	Backup & Recovery	Important application data should be backed up regularly and recoverable after system failure.
NFR15	AI Reliability	AI-generated suggestions should be treated as recommendations and should be reviewable by authorized users.
NFR16	AI Performance	AI-assisted classification should return results within an acceptable time during normal system operation.

6. System Modules

Module	Main Functions
Authentication Module	User, Staff, and Admin login; JWT authentication; role-based access
User Management Module	Registration, login, profile, account management
Staff Management Module	Staff accounts, profiles, department assignment, account management
Request Management Module	Request creation, details, attachments, location, status
Task Management Module	Task creation, assignment, deadlines, progress, completion
Search & Filter Module	Search and filter requests by category, status, priority, department, and date
Assignment Module	Request assignment and reassignment to staff
Resolution Module	Status updates, remarks, resolution evidence, request history
AI Assistance Module	Request classification, department recommendation, priority prediction, duplicate detection
Notification Module	Request submission, assignment, status, task, and resolution notifications
Feedback Module	User feedback and service ratings
Reports Module	Request, category, department, priority, staff, and resolution reports
Dashboard & Analytics Module	Operational statistics, request trends, resolution rate, staff performance
Administration Module	User, staff, department, category, and system management
Audit Module	User activity monitoring and administrative audit logs

7. Sample Use Cases

Use Case 1: User Creates a Service Request

Field	Description
Use Case ID	UC-01
Use Case Name	Create Service Request
Primary Actor	User / Employee
Goal	Allow a user to submit a service request with relevant details.
Preconditions	User has a registered active account and is logged in.

Main Flow

1. User logs in.
2. Opens the **Create Request** page.
3. Enters the request title.
4. Enters the request description.
5. Selects a request category or allows AI to suggest one.
6. Uploads supporting images/documents if required.
7. Provides relevant request information.
8. System validates the request.
9. AI analyzes the request.
10. AI suggests the category and priority.
11. System records the request.
12. System generates a request reference number.
13. Request becomes available to the Administrator.
14. User receives a request-submission notification.

Alternative Flow

If required information is missing, the system displays an appropriate validation message and asks the user to complete the missing information.

Postconditions

The request is recorded in the system and becomes available for administrative processing.

Related Requirements

FR-U10–FR-U15, FR-AI01–FR-AI05

Use Case 2: Administrator Assigns a Request to Staff

Field	Description
Use Case ID	UC-02
Use Case Name	Assign Service Request to Staff
Primary Actor	Administrator
Goal	Allow the Administrator to review a request and assign it to the appropriate staff member.
Preconditions	Administrator is logged in and a request has been submitted.

Main Flow

1. Administrator logs in.
2. Opens the Admin Dashboard.
3. Views pending requests.
4. Selects a request.
5. Views request details.
6. Reviews description and attachments.
7. Reviews category and priority.
8. Reviews AI-generated category and priority.
9. Selects the appropriate department.
10. Selects an available staff member.
11. Assigns the request.
12. System records the assignment.
13. Staff receives a notification.
14. User receives an assignment notification.

Alternative Flow

If no suitable staff member is available, the Administrator can keep the request pending or assign it later.

Postconditions

The request is assigned to a staff member and the assignment is recorded in the system.

Related Requirements

FR-A09–FR-A14, FR-AI08–FR-AI09

Use Case 3: Staff Resolves a Service Request

Field	Description
Use Case ID	UC-03
Use Case Name	Resolve Service Request
Primary Actor	Staff
Goal	Allow staff to process an assigned request and mark it as resolved.
Preconditions	Staff account exists, staff is logged in, and a request has been assigned to the staff member.

Main Flow

1. Staff logs in.
2. Opens the Staff Dashboard.
3. Views assigned requests.
4. Selects a request.
5. Views request details.
6. Accepts the request.
7. Updates the status to **In Progress**.
8. Performs the required service activity.
9. Adds resolution remarks.
10. Uploads resolution evidence if required.
11. Updates the status to **Resolved**.
12. System records the resolution information.
13. User receives a resolution notification.
14. User can provide feedback.

Alternative Flow

If the issue cannot be resolved immediately, the staff member can keep the request **In Progress** and add appropriate remarks.

Postconditions

The request status is updated and the resolution information is stored.

Related Requirements

FR-ST03–FR-ST14, FR-U18–FR-U26

Use Case 4: AI Detects a Duplicate Request

Field	Description
Use Case ID	UC-04
Use Case Name	Detect Potential Duplicate Request
Primary Actor	System / AI
Goal	Identify whether a newly submitted request may already exist in the system.
Preconditions	A new request has been submitted and existing requests are available.

Main Flow

1. User submits a request.
2. System stores the request information.
3. AI analyzes the request description and relevant information.
4. AI compares it with existing requests.
5. AI calculates a similarity score.
6. System identifies potentially similar requests.
7. System displays possible duplicate requests to the Administrator.
8. Administrator reviews the suggestions.
9. Administrator confirms or rejects the duplicate suggestion.
10. System records the Administrator's decision.

Alternative Flow

If no similar request is detected, the request continues through the normal request-management workflow.

Postconditions

Potential duplicate information is recorded and available for Administrator review.

Related Requirements

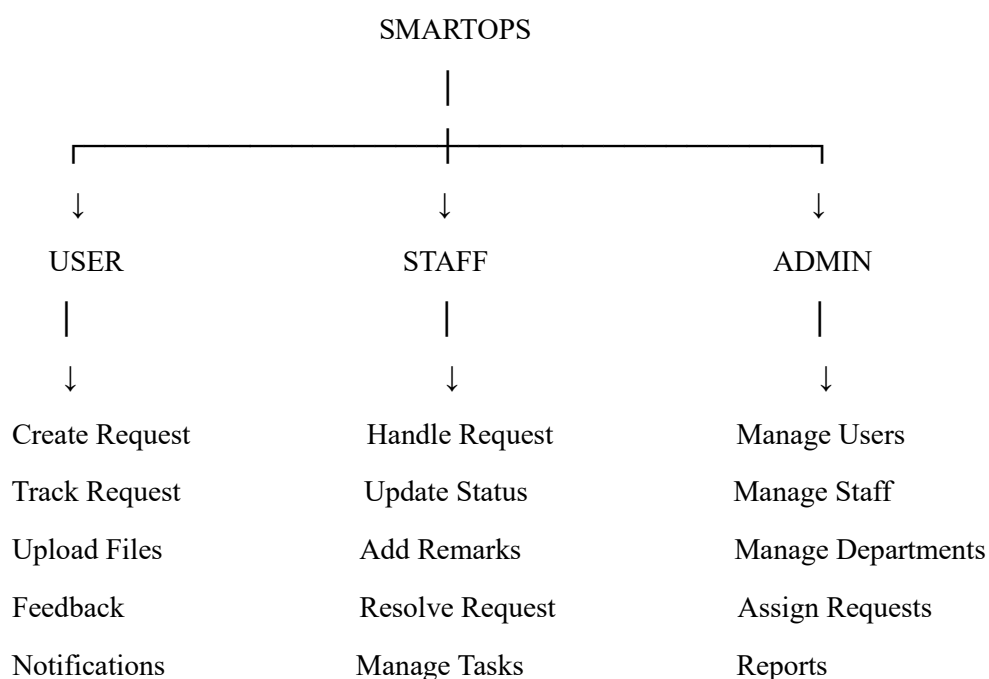
FR-AI06–FR-AI10, FR-A19

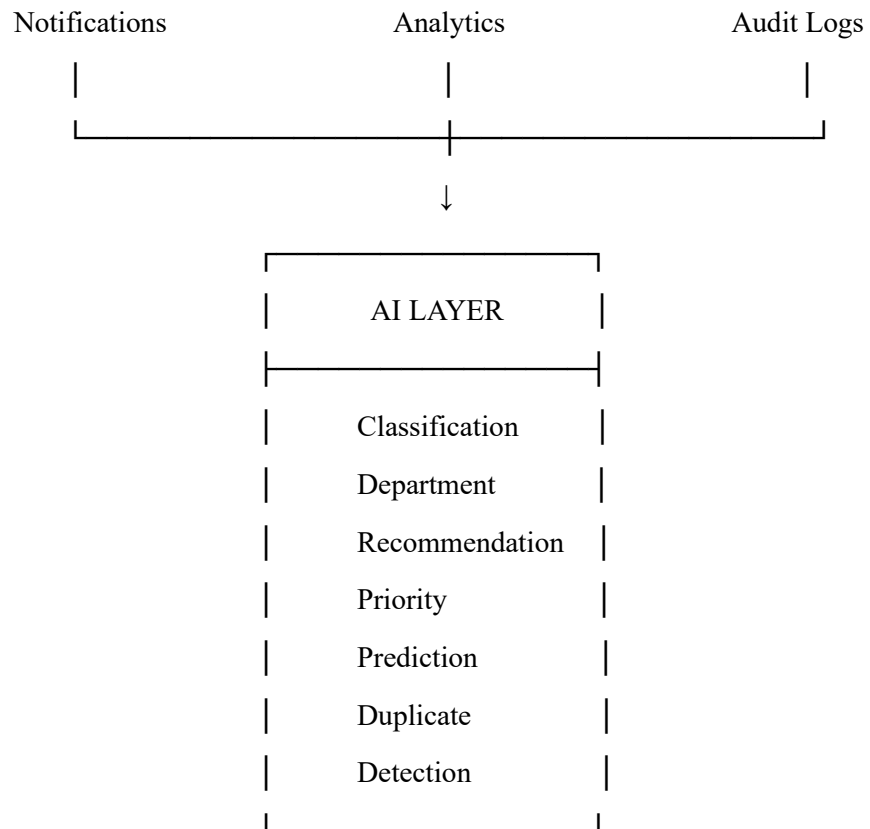
8. Assumptions and Clarifications

- The authentication mechanism will use a secure authentication approach such as JWT-based authentication.
- Passwords will be securely hashed before being stored.
- The exact AI model/provider has not been specified.
- The exact AI classification algorithm has not been specified.

- The exact AI confidence threshold for category and priority prediction has not been specified.
- AI-generated category and priority will be treated as recommendations and can be reviewed by authorized Administrators.
- The exact file-storage mechanism has not been specified.
- The exact notification method, such as email, SMS, push notification, or in-system notification, has not been specified.
- Exact request categories will be configured by the Administrator.
- Exact department structure will be configured by the Administrator.
- Exact request priority rules have not been fully specified.
- Exact request resolution time/SLA has not been specified.
- Exact duplicate-detection threshold has not been specified.
- Exact feedback and rating calculation method has not been specified.
- Exact format of administrative reports has not been specified.
- Exact performance, availability, and security targets have not been specified.
- Exact password-reset mechanism has not been specified.
- Payment processing is not included in the current scope.
- AI does not independently make final administrative decisions; authorized users can review AI suggestions.
- The application is intended to support organizational operations and request management rather than replace the decision-making authority of managers or administrators.

9. Project Scope at a Glance





Final project definition

SmartOps is a MERN-based organizational operations and service request management system enhanced with AI-assisted request classification, department recommendation, priority prediction, and duplicate request detection.